Post-Quantum Cryptography Migration

Part 16 - Edges & Sectors: Enterprise IoT & OT

Compiled by the Spaced Research Ledger

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The physical world runs on devices that live for decades: grid controllers, payment cards, cars, satellites, smart meters. They are the hardest things to migrate and the last to move, which makes this report a map of gaps as much as progress.

Eleven sections cover the terrain. Sections 1 and 2 cover industrial protocols and the energy grid. Sections 3 through 5 cover physical security, medical devices, and payments hardware. Sections 6 and 7 cover vehicles, aviation, and rail. Sections 8 and 9 cover short-range wireless and IoT messaging, and Sections 10 and 11 cover telecom and satellites.

Where the sectors have moved, it is at the standards and silicon level. The power industry's DNP3 protocol grew a post-quantum security layer, smart metering's DLMS standard is finalizing ML-KEM and ML-DSA, and automotive security chips with the algorithms in hardware are shipping. Payment-card feasibility is settled: only ML-KEM and ML-DSA fit on card hardware at all.

Where the sectors have not moved, the physics is the obstacle. A single ML-KEM handshake can fragment into 62 LoRa packets, vehicle-to-vehicle messages cannot fit post-quantum signatures inside their packet limits, and Matter and Thread have no post-quantum statement at all. The clocks are running anyway, led by France's 2027 certification cutoff and the EU's 2030 critical-infrastructure deadline.

1. Industrial Control Protocols

Factories, water plants, and substations speak protocols designed decades before security was a requirement, let alone quantum resistance. Retrofits arrive as security layers bolted around the old messages. This section covers the industrial protocol families, where one real post-quantum layer now exists and the rest is research and committee work.

Current State

The power industry produced the concrete artifact. DNP3-SL, the protocol's new security layer, uses post-quantum cryptography including ML-KEM for establishing mutually authenticated associations, with symmetric sessions built for constrained brownfield devices [1]. It is designed as a separable layer, supports PKI or self-signed certificates, eliminates preshared secrets, and coexists with the prior security version [2]. The history explains the caution, with earlier classical retrofits criticized as incomplete [3].

The rest is profile work and proposals. IEC 62351, the power-grid security standard family, has begun its transition with a cryptographic inventory and working-group effort, using its migration-guidance part for stronger algorithms [4]. Several parts profile standard protocols like TLS, so those move by profile update, while others need dedicated committee work [4]. Key-management updates for the multicast grid protocols are under consideration [5], and summaries acknowledge the quantum vulnerability without specifying implementations [6].

Modbus has no security in its base specification at all [7], with research proposing proxy-based post-quantum frameworks for legacy devices [7] and measured integrations into its secure variant [8]. OPC UA has peer-reviewed hybrid proposals but no vendor-shipped integration [9], continued survey attention [10], and a European consortium proposal to build the real thing [11]. Building automation's secure profile rides TLS 1.3 and certificates, with the IETF noting both must migrate [12].

2. Energy & Grid

The electric grid is critical infrastructure with equipment lifecycles measured in decades and, so far, no binding post-quantum rules. Regulators are circling, vendors are partnering, and the meters on houses are becoming the first mass deployment question. This section covers the grid's regulatory vacuum and the smart-meter supply chain filling it.

Recent Developments

The rules are still forming. No U.S. post-quantum regulation exists for energy, and no state has quantum-specific requirements [13]. The proposed Quantum-GUARD Act would push grid regulators to weigh quantum risk and fund a testing environment, but remains introduced, not enacted [14][15]. The June executive order commits federal assistance rather than mandates [15].

Abroad the clocks are set: the EU targets critical-infrastructure quantum resistance by 2030, the UK phases to 2035 [13], and France stops certifying products without quantum-resistant encryption in 2027 [16]. China, meanwhile, deployed dozens of quantum devices at a demonstration substation [17]. On meters, Honeywell is integrating quantum-generated keys with both classical and post-quantum algorithms [18], and the telecom industry is testing the NIST algorithms on constrained meters [19]. Open-standard mesh architectures are argued to absorb the algorithms more easily than proprietary stacks [20].

Current State

The binding-rules gap is explicit. NERC's roadmap names quantum computing an emerging risk with no requirements attached [21], and the proposed legislation would create testing rather than mandates [22]. Australia's guidance asks energy operators for transition plans during 2026 [23].

The meter ecosystem is organizing around standards and silicon. The DLMS metering-security standard is finalizing its Suite 3 with ML-KEM and ML-DSA [24], requiring a post-quantum-ready secure element with physical isolation and secure update [24]. SEALSQ partnered with the dominant mesh-networking alliance to bring post-quantum PKI and hardware to metering [25], with its secure element's certifications tracking toward year end [26]. PKI tooling now targets replacing the identical factory-burned symmetric keys in meter fleets [27].

The stakes are longevity: meters installed today run into the 2040s, making agility the priority over algorithm perfection [28]. A market forecast sees quantum-resistant meters tripling by 2034 [29], while a systematic review finds deployed edge mitigations entirely unprepared, a gap between academic proposals and shippable hardware [30].

3. Physical Security Devices

The cameras and access controllers watching buildings are networked computers with certificate stores, standardized by ONVIF. Nothing in that standard is post-quantum yet, and the vendors are at the considering stage. This section is short because the sector is early.

Current State

The standard enumerates only classical algorithms. ONVIF's current security specification lists RSA and ECC key pairs for its keystore, with no post-quantum algorithm named anywhere [31]. The structure for change exists: security moved into an add-on category precisely so it can update faster than fixed profiles [32]. The add-on is slated for upgrade at the end of 2026 against new cybersecurity requirements [33].

The standard's scope spans cameras and access control across tens of thousands of conformant products [33]. On the vendor side, Axis says post-quantum considerations now enter design decisions across its product lines, alongside FIPS work [34].

4. Medical Devices

Medical devices combine decades-long lifecycles, patient-safety constraints, and some of the most power-starved electronics anywhere. The regulator's cryptography requirements remain general rather than quantum-specific, while research produces hardware that could actually run the algorithms on a pacemaker's power budget. This section covers both tracks.

Recent Developments

The quality regime tightened without naming quantum. The FDA's Quality Management System Regulation took effect in February 2026, embedding cybersecurity into design controls and post-market surveillance as a mandatory quality element [35].

Current State

The regulatory posture is general-purpose. The premarket guidance requires lifecycle protection assurance, citing general key-management standards, nothing post-quantum-specific [36]. Its 2026 reissue was a citation alignment, core requirements unchanged [37]. A MITRE discussion paper covers post-quantum risk analysis for manufacturers, influential but non-binding [38]. A systematic review maps the frameworks, recommends staged migration to protect patient safety, and names pervasive legacy systems as the structural barrier [39].

The hardware research is the bright spot. An MIT chip the size of a needle tip runs the post-quantum algorithms on wireless biomedical power budgets, an order of magnitude more efficient with anti-tampering built in [40]. The architectural alternative offloads the work, with a constrained device doing one lightweight exchange to a cloud broker at session start [41].

5. Payments Hardware

Every card tap runs cryptography on some of the most constrained silicon in commerce, governed by EMV standards built around RSA. The feasibility mathematics is now settled, the chips are entering production, and the standards body says there is no hurry. This section covers that tension.

Recent Developments

The hardware is arriving ahead of the standard. Diebold Nixdorf designs its current ATM lines for post-quantum readiness [42], noting machines bought in 2026 serve into the mid-2030s, exactly one refresh cycle against the federal deadlines [43]. IDEMIA demonstrated KEM-based authentication replacing signatures on a smart card [44]. SEALSQ's secure microcontroller is entering transit payment readers [45], and STMicroelectronics's mobile chip targets certification with an ML-KEM and ML-DSA accelerator [46].

NXP published its transition framework for secure elements through 2030 [47]. One widely repeated claim needs a caveat: the reported Visa and Mastercard phased upgrade through 2028 traces to a single uncorroborated commentary, not a network announcement [48].

Current State

The feasibility line is drawn precisely. IDEMIA's analysis found most candidates simply do not fit on card hardware, with only ML-KEM and ML-DSA theoretically feasible, and side-channel protection multiplying execution time up to five-fold [49]. Its earlier hybrid implementation showed EMV's 256-byte command limit too small for post-quantum signatures, requiring extended commands [49]. Its chip partnership targets mass production in 2026 [50], following its first post-quantum hardware accelerator [51].

The standards body counsels patience while the sector plans anyway. EMVCo expects no practical quantum threat to EMV before 2040, possibly never, and is monitoring with industry engagement planned [52], publishing its position-paper series [53]. Financial-sector guidance is blunter, noting it is not known whether EMV can even accommodate quantum-safe certificates [50]. The sector's working group has nonetheless reviewed ATM and POS vulnerabilities [54].

ATM security leadership names quantum among coming threats [55]. Compact libraries fit the algorithms into terminal memory budgets [56], and a quantum-safe QR terminal collaboration is underway [57]. Tokenization helps fraud but does not touch the underlying asymmetric exposure [58].

6. Automotive

A car is a datacenter on wheels with a 15-year life and a type-approval regime that makes cryptographic change slow. The chips arrived first, the software standards are agile but algorithm-silent, and the vehicle-to-vehicle radio problem is truly unsolved. This section covers all three layers.

Recent Developments

Production silicon crossed the line. Telechips ships automotive processors with a security engine implementing all three NIST standards in hardware, isolated across cockpit and driver-assistance domains [59]. NXP reports major automakers already implementing post-quantum in software-defined vehicle platforms, with production vehicles expected within a couple of years [60]. The tooling followed: IAV's quantumSAR library covers the NIST set plus the backup KEM against the current AUTOSAR release [61], and architecture proposals sketch a quantum-safe crypto abstraction layer [62].

Korea funded a quantum-resistant automotive security chip program running to 2030, tied to the type-approval standards [63]. Industry timing guidance targets high-risk use cases before 2030, with broad deployment by 2035 under vehicle-lifecycle constraints [64].

Current State

The software standard is agile but uncommitted. AUTOSAR's crypto stack is modular by construction, with provider interfaces and KEM abstractions that avoid locking algorithms in [65], yet no post-quantum schemes are specified for compliance [66]. IAV's design philosophy fills the gap in software, replacing crypto drivers without replacing the ECU [67].

Regulation frames the deadline without naming algorithms. The UNECE cybersecurity and update regulations are mandatory for type approval, blocking sales without certified management systems [68], mapped through the ISO engineering standards [68]. Migration is practical risk management rather than legal mandate, with 2026 to 2028 the window before hardware lead times narrow options [68]. Deprecated algorithms may risk type approval [69].

Roughly half a vehicle's ECUs are security-relevant [70], and the field has its own academic survey [71]. A vendor push ties safety and security compliance together [72].

Vehicle-to-vehicle is the hard residue. Current V2X security is entirely elliptic-curve, and its certificate format has no post-quantum forward compatibility [73]. That creates a deployment stalemate where introducing new certificates degrades both old and new vehicles [73]. The packet arithmetic is brutal, with certificate limits in the low hundreds of bytes against multi-kilobyte signatures [74].

The responses are inventive. A hybrid certificate scheme cuts pseudonym certificates under the packet limit [75], and certificate segmentation targets cellular V2X [76]. A credential system replaces per-message signatures with zero-knowledge rate limiting [77], and a Korean signature algorithm benchmarks well [78]. Structured migration roadmaps run to 2038 [79], a protocol proposal covers 5G-V2X authentication [80], and a review catalogues thirteen open gaps [81].

7. Aviation & Rail

Aircraft and trains run communication systems certified over decades, some still unencrypted entirely. Both sectors have started the paperwork phase of the quantum transition, with the American airspace regulator asking industry how and Europe's rail programs writing agility into new specifications. This section covers two industries at the request-for-information stage.

Current State

Aviation's baseline is startling. Aircraft voice and the ACARS datalink are largely unencrypted and unauthenticated in practice, with 99 percent of ACARS traffic in plain text [82]. The surveillance broadcast carries no cryptography at all [82]. The FAA issued a formal request for input on transitioning the airspace system and its enterprise IT to post-quantum, framed as prerequisite to the coming decades [83].

It extended the deadline once [84]. The governing frameworks are its cybersecurity strategy and ICAO's action plan [85].

Vendors market drop-in protection for the legacy channels, unverified beyond marketing [85]. Research adapts avionics certification itself, wrapping hybrid post-quantum encryption inside module-isolation boundaries [82], and a European doctoral project targets quantum-resistant protocols for air-traffic communications [86].

Rail is riding its generational upgrade. The industry named quantum security a need as early as 2020, before the algorithms were ready [87]. The legacy GSM-R radio is being replaced for 2G obsolescence, not quantum, on a 2030 clock [88], with the 5G successor in trials toward a 2027 specification [89]. Europe's rail cybersecurity specification now explicitly designs an open architecture ready to evolve toward post-quantum measures [90].

Vendor practice puts quantum-safe symmetric encryption at the transport layers first, with post-quantum algorithms later [91][92]. The research layer ties it to railway safety standards, notes signaling assets running into the 2040s, and urges pilots now [93]. Systematic analyses recommend sector PKI and explicit agility [94].

8. Short-Range IoT

Bluetooth, Zigbee, Thread, and Matter connect the smallest devices in the building, radios with payloads measured in dozens of bytes. Post-quantum keys do not fit nicely, and the standards bodies have said nothing. This section covers a gap documented mostly by academics measuring exactly how badly things fragment.

Recent Developments

The measurements arrived. Every NIST parameter set runs on a common microcontroller without external memory, latency and footprint quantified [95]. The fragmentation study puts numbers on the radio problem, up to 62 packets for one ML-KEM handshake on low-payload LoRa configurations, with Bluetooth and 802.15.4 in between [96]. The standards silence is itself a finding: nothing from the Matter or Thread organizations addresses post-quantum at all, through four specification releases [97].

Current State

The community noticed before the standards did. Smart-home forums flag that Zigbee and Matter-over-Thread security is classical with most devices lacking hardware ever to upgrade [98]. The research is filling in. ML-KEM is computationally feasible in Bluetooth Classic pairing on common cores while the backup KEM is not, at the cost of airtime and loss sensitivity [99].

Energy studies find communication overhead dominating computation [100], and a MILCOM paper ran ML-KEM inside a Zigbee stack [101]. Compact non-standard KEMs trade size against standardization [102], and a review calls Bluetooth post-quantum research still limited [103].

9. IoT Messaging & LPWAN

Above the radios sit the messaging protocols, MQTT and CoAP, and the long-range networks like LoRaWAN. One of these was accidentally quantum-resistant all along. This section covers the measured cost of post-quantum messaging and the network that never used public-key cryptography in the first place.

Recent Developments

MQTT got its hybrid proposals and its price tag. A hybrid protocol pairs ML-KEM with X25519 for the publish-subscribe world [104]. A measured migration on emulated IoT hardware found handshakes nearly five times slower under network impairment while steady-state publishing stays negligible [105]. Persistent sessions are the recommended mitigation [105].

Current State

LoRaWAN is the exception that proves the rule. Its security has always been purely symmetric, two preshared root keys deriving sessions, no public-key exchange at all [106]. That means no Shor exposure, but static never-rotated keys are a documented weakness, and the roadmap answer is stronger symmetric cryptography rather than post-quantum [107]. Research proposes post-quantum root-key renewal using isogeny-based exchange with a minimal PKI [108]. A dedicated key-management design project is underway [109], and earlier work bolted a lattice KEM on directly [110].

The messaging measurements are maturing into scores. A study across MQTT, HTTP, and HTTPS under real post-quantum load on constrained hardware built a composite readiness score [111]. MQTT was most efficient by architecture, nothing reached excellent readiness, and distance compounds the cryptographic cost [111][111]. The field predates it by years [111].

CoAP research found its own gap, IoT-specific protocols neglected relative to TLS [112]. It then delivered a surprise: verified post-quantum handshakes on a microcontroller completing 63 percent faster than the classical baseline [113]. Signature work runs Falcon over MQTT on real boards [114], and an 8-bit sensor-node KEM supports hybrid messaging where nothing else fits [115].

10. Telecom & 5G

Mobile networks guard subscriber identity with elliptic curves and run roaming interfaces that make attractive interception points. The standards machinery has formally started its post-quantum study, while operators run the world's earliest SIM-card demonstrations. This section covers a sector migrating by committee, with the committee finally in session.

Recent Developments

The formal study exists. 3GPP's security group created its post-quantum transition study for Release 20, under change control since June 2026, covering the identity-concealment protocol among others [116].

Current State

The machinery is aligned even if the mandates are not. The security group's work was triggered by the telecom association's guidelines and European recommendations, producing a catalogue of cryptography across the 5G specifications [117]. With Release 19 complete, post-quantum is named the next step alongside 6G security [118]. Commentary expects quantum-resistant options for authentication and identity encryption in coming releases, none formally adopted yet [119].

The association's task force spans dozens of operators [119] and extended its guidance to connected IoT [120]. It mapped the roaming interfaces as the common attack entry, with a quantum-safe upgrade path prescribed [121]. The near-term posture is agility and inventories over forklift swaps [122]. The research says the costs are tolerable, with core-network algorithm swaps showing small effects [123], and an open-source quantum-secure 5G core exists [124].

Subscriber identity is the concrete vulnerability. The concealment scheme's two profiles are entirely elliptic-curve, fully exposed to a quantum adversary [125]. Device lifecycles beyond ten years drive the urgency, with decade-old SIMs still deployed as precedent [126]. Research replaces the scheme with ML-KEM in the authentication flow [125], hybrid protocol proposals exist [127], and the work extends into 6G [128]. Operators demonstrated it live years ago, with quantum-resistant SIMs concealing identities on a real network [129] and hybrid encryption piloted on live traffic [129].

11. GNSS & Satellite Links

Satellites broadcast the timing that underpins finance and the navigation that underpins everything else, signed with classical curves where signed at all. Space systems cannot be recalled for upgrades, and the standards body for space links has no post-quantum extension. This section covers the orbit-height view of the migration's furthest frontier.

Recent Developments

The standards gap is the headline. No ratified satellite-navigation authentication standard specifies a post-quantum signature, with every deployed scheme classical [130]. Galileo's ranging-level authentication service entered testing [131]. The commercial layer is moving anyway, with a post-quantum satellite constellation's first launch scheduled [132] and manufacturing partnerships extending it to India [133]. A systematic survey names incomplete space-standards work the top research gap, proposing a phased roadmap to 2040 [134].

Current State

Navigation authentication went operational, classically. Galileo's OSNMA reached initial service in July 2025, the first global open-service signal authentication [135]. It embeds signatures in the navigation message [136], built on a delayed-key broadcast scheme in 40 spare bits [130]. Verification today runs against elliptic-curve keys and a Merkle tree [137], with accumulation requirements [137] and tested key-rollover machinery [138]. It detects spoofing rather than preventing it [139].

The post-quantum path is charted: a European project validated hash-based structures achieving quantum resilience at low latency, plus Merkle-tree architectures needing further study [140][140]. Other constellations' research marks post-quantum selection as open [141][142], and augmentation-system assessments remain incomplete [143].

The command links are symmetric and stalled. The space data-link security standard provides authenticated encryption over preshared symmetric keys, no public-key mechanism at all [144][145][146], across its documented portfolio [147], with management extensions added in 2020 [148]. No post-quantum extension is ratified, with hybrid ML-KEM and ML-DSA extensions remaining research [149], and even the existing standard's adoption has been slow [148].

The demonstrations are outrunning the paperwork. A European hybrid extension ran on constrained hardware with FPGA acceleration [150], and an agency project is building post-quantum satcom hardware [151]. A lightweight framework targets satellite 5G [152].

The first post-quantum link from the International Space Station used a proxy requiring no code changes [153], and a Starlink link carried post-quantum traffic in a claimed U.S. first [154]. Validated overlay products target satcom without hardware refresh [155], and nanosatellite-scale encryption has been prototyped [156]. The harvest logic drives it all, since telemetry and imagery hold value for decades [155].

About This Report

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